Control variates cancel the leading client-heterogeneity drift in a federated two-step round

Companion proof for: pleyva2004/communication-efficient-learning-of-deep-networks
Improvement slug: `heterogeneity-control-variate`

Abstract

The math deep dive (eq. (3.5)) showed that running two local gradient-descent steps per client and then averaging deviates from two genuine centralized gradient-descent steps by a leading term $\eta^2 \operatorname{Cov}_k(H_k, g_k) + O(\eta^3)$, the client-weighted covariance between local Hessians H_k and local gradients g_k. This is the "client-drift" object that later federated-learning theory (e.g. Karimireddy et al. 2020, SCAFFOLD) is explicitly designed to remove. Here we prove, by carrying the *same* Taylor expansion but replacing each local gradient $g_{\text{local}}(w)$ with the control-variate-corrected direction $g_{\text{local}}(w) - c_k + c$ (where c_k is a fixed estimate of client k's gradient at the shared point w_t and $c = \sum_k \alpha_k c_k$ is its client-weighted mean), that the entire order-η^2 covariance term cancels. Consequently the corrected averaged two-step round matches two centralized gradient-descent steps up to $O(\eta^3)$, recovering, to leading order, the per-round progress of centralized full-batch gradient descent even on an arbitrarily heterogeneous (non-IID) partition.

Setup and notation

We adopt verbatim the setup of the math deep dive. The global objective is the finite sum $f(w) = \sum_{k=1}^{K} \alpha_k F_k(w)$ with mixture weights $\alpha_k := n_k/n \geq 0$, $\sum_{k=1}^{K} \alpha_k = 1$, where $F_k(w) = \frac{1}{n_k} \sum_{i \in P_k} f_i(w)$ is client k's local objective over its example set P_k of size n_k; this is the exact algebraic identity (1.1) of the deep dive, valid for any partition. All clients begin a round at the common shared point $w_t \in \mathbb{R}^d$ (the load-bearing shared-initialization assumption). We fix the round and write everything relative to w_t.

We work in the *quadratic / second-order local model* used to derive eq. (3.5): each F_k is twice continuously differentiable in a neighborhood of w_t, and we expand each local gradient to first order about w_t with a uniform $O(\cdot)$ remainder. Concretely, writing $g_k := \nabla F_k(w_t)$ and $H_k := \nabla^2 F_k(w_t)$, for any displacement δ of size $O(\eta)$ Taylor's theorem with remainder gives

$$\nabla F_k(w_t + \delta) = g_k + H_k \, \delta + R_k(\delta), \qquad \|R_k(\delta)\| = O(\|\delta\|^2), \tag{1}$$

where the constant hidden in $O(\|\delta\|^2)$ is uniform over the finitely many clients $k \in \{1, \dots, K\}$ (take the maximum over k of the per-client constants). Because every displacement we encounter is $O(\eta)$, every such remainder contributes at order η^2 *inside* a term that is already multiplied by η, i.e. at order η^3 in the position iterate; this is exactly the bookkeeping that makes the deep dive's $O(\eta^3)$ statement precise. The learning rate is $\eta > 0$, taken small enough that the round's displacements stay in the neighborhood where (1) holds.

Definition 1 (Working notation)**.** Throughout the round:

- $g_k := \nabla F_k(w_t) \in \mathbb{R}^d$ — client k's local gradient at the shared point w_t.

- $H_k := \nabla^2 F_k(w_t) \in \mathbb{R}^{d\times d}$ — client k's local Hessian at w_t.
- $\nabla f := \nabla f(w_t) = \sum_{k=1}^{K} \alpha_k g_k$ — the global gradient at w_t; the equality is the exact gradient-aggregation identity (2.2) of the deep dive, obtained by differentiating $f = \sum_k \alpha_k F_k$.
- $H := \nabla^2 f(w_t) = \sum_{k=1}^{K} \alpha_k H_k$ — the global Hessian at w_t, again by differentiating the mixture identity twice.
- $\mathrm{Cov}_k(H_k, g_k) := \sum_{k=1}^{K} \alpha_k H_k g_k - \Big(\sum_{k=1}^{K} \alpha_k H_k\Big)\Big(\sum_{k=1}^{K} \alpha_k g_k\Big) = \sum_{k=1}^{K} \alpha_k H_k g_k - H\,\nabla f$ — the client-weighted covariance between local Hessians and local gradients (a vector in $\mathbb{R}^d$), with the α_k acting as the client distribution. This is the leading deviation term of eq. (3.5).
- **Control variates.** Each client k carries a fixed vector c_k that estimates its own local gradient direction at the shared point w_t; in the static, exact-estimate regime analyzed here we take $c_k := g_k = \nabla F_k(w_t)$, held *constant* for the whole round (it is computed once at w_t and not refreshed between the two local steps). The server control variate is the client-weighted mean $c := \sum_{k=1}^{K} \alpha_k c_k = \sum_{k=1}^{K} \alpha_k g_k = \nabla f$, also held constant for the round.
- **Corrected local direction.** At any iterate w, instead of stepping along the raw local gradient $g_{\mathrm{local}}(w) := \nabla F_k(w)$, client k steps along the control-variate-corrected direction

$$\tilde{g}_k(w) := \nabla F_k(w) - c_k + c. \tag{2}$$

Theorem

Theorem 2 (Control variates cancel the order-η^2 heterogeneity drift)**.** *Fix a round at the shared point w_t and adopt the quadratic / second-order local model (1) with static, exact control variates $c_k = g_k$ and $c = \sum_k \alpha_k g_k = \nabla f$. Let each client run* two *local steps using the corrected direction (2),*

$$w^k_{(0)} = w_t, \qquad w^k_{(1)} = w^k_{(0)} - \eta\,\tilde{g}_k(w^k_{(0)}), \qquad w^k_{(2)} = w^k_{(1)} - \eta\,\tilde{g}_k(w^k_{(1)}),$$

and let the server form the client-weighted average $\bar{w}_{t+2} := \sum_{k=1}^{K} \alpha_k w^k_{(2)}$. Let $G^{(2)}(w_t)$ denote two genuine centralized gradient-descent steps on f from w_t, namely $u_1 = w_t - \eta\,\nabla f$, $G^{(2)}(w_t) = u_1 - \eta\,\nabla f(u_1)$. Then the order-$\eta^2$ deviation $\eta^2\,\mathrm{Cov}_k(H_k, g_k)$ of eq. (3.5) vanishes to first order: the corrected averaged round equals two centralized steps up to $O(\eta^3)$,

$$\bar{w}_{t+2} = w_t - 2\eta\,\nabla f + \eta^2 H\,\nabla f + O(\eta^3) = G^{(2)}(w_t) + O(\eta^3),$$

whereas the uncorrected *FedAvg round (using g_{local} directly) satisfies $\bar{w}^{\mathrm{FedAvg}}_{t+2} = G^{(2)}(w_t) + \eta^2\,\mathrm{Cov}_k(H_k, g_k) + O(\eta^3)$. In particular the leading discrepancy is reduced from $\Theta(\eta^2)$ (generically nonzero under client heterogeneity) to $O(\eta^3)$.*

Proof

Proof. The proof carries the same Taylor expansion as the deep dive's derivation of eq. (3.5), but applied to the corrected direction (2). We proceed in five steps: (i) compute the corrected first step; (ii) Taylor-expand the corrected second step; (iii) assemble the corrected two-step endpoint per client; (iv) average over clients; (v) compare against two centralized steps and against the uncorrected FedAvg round, exhibiting the cancellation of the covariance term.

Step (i): the corrected first step is identical across clients and equals one centralized step. Evaluate (2) at the shared start $w^k_{(0)} = w_t$. By definition $\nabla F_k(w_t) = g_k$ and, in the static exact-estimate regime, $c_k = g_k$, so the raw local gradient and the client's own control variate cancel exactly:

$$\tilde{g}_k(w_t) = \underbrace{\nabla F_k(w_t)}_{=g_k} - \underbrace{c_k}_{=g_k} + c = g_k - g_k + c = c = \nabla f. \tag{3}$$

This is the crux of the correction: the term $-c_k$ removes the client's own gradient and $+c$ substitutes the *global* direction ∇f, so the first corrected step is the same for every client and equals exactly one centralized gradient-descent step,

$$w^k_{(1)} = w_t - \eta\, \tilde{g}_k(w_t) = w_t - \eta\, \nabla f \qquad \text{for every } k. \tag{4}$$

In particular $w^k_{(1)}$ does not depend on k; we may write it as $u_1 := w_t - \eta\, \nabla f$, which is precisely the first centralized iterate.

Step (ii): Taylor-expand the corrected second direction. We now evaluate (2) at $w^k_{(1)} = w_t - \eta\, \nabla f$. The displacement from w_t is $\delta := -\eta\, \nabla f$, which is $O(\eta)$, so (1) applies with remainder $\|R_k(\delta)\| = O(\eta^2)$:

$$\nabla F_k(w_t - \eta\, \nabla f) = g_k + H_k(-\eta\, \nabla f) + R_k(-\eta\, \nabla f) = g_k - \eta\, H_k\, \nabla f + O(\eta^2). \tag{5}$$

Subtract the static control variate $c_k = g_k$ and add $c = \nabla f$. The constant local gradient g_k cancels against $-c_k = -g_k$ a *second* time — this is the structural reason the correction keeps working past the first step — leaving

$$\tilde{g}_k(w^k_{(1)}) = \big(g_k - \eta\, H_k\, \nabla f + O(\eta^2)\big) - g_k + \nabla f = \nabla f - \eta\, H_k\, \nabla f + O(\eta^2). \tag{6}$$

Notice that the leading $O(1)$ part of the corrected second direction is again the global ∇f, with the *only* client-dependent piece appearing at order η through the client's own Hessian H_k acting on the *global* gradient ∇f — not on the local g_k. This single change (the matrix H_k now multiplies ∇f rather than g_k) is what will make the covariance collapse in Step (v).

Step (iii): assemble the corrected two-step endpoint per client. Combine (4) and (6):

$$\begin{aligned} w^k_{(2)} &= w^k_{(1)} - \eta\, \tilde{g}_k(w^k_{(1)}) \\ &= (w_t - \eta\, \nabla f) - \eta(\nabla f - \eta\, H_k\, \nabla f + O(\eta^2)) \\ &= w_t - \eta\, \nabla f - \eta\, \nabla f + \eta^2 H_k\, \nabla f + \eta \cdot O(\eta^2) \\ &= w_t - 2\eta\, \nabla f + \eta^2 H_k\, \nabla f + O(\eta^3). \end{aligned} \tag{7}$$

Here the term $\eta \cdot O(\eta^2) = O(\eta^3)$ absorbs the propagated Taylor remainder: the $O(\eta^2)$ error in the second direction (6) is multiplied by the step size η, hence enters the position iterate (7) only at order η^3, exactly as claimed. This is the corrected analogue of the deep dive's uncorrected endpoint $w^k_{(2),\mathrm{FedAvg}} = w_t - 2\eta g_k + \eta^2 H_k g_k + O(\eta^3)$: the $O(1)$ term has moved from the client-specific $-2\eta g_k$ to the global $-2\eta\, \nabla f$, and the $O(\eta^2)$ term from $H_k g_k$ to $H_k\, \nabla f$.

Step (iv): average over clients. Apply the server's client-weighted average $\sum_k \alpha_k(\cdot)$ to (7).

Linearity of the sum and $\sum_k \alpha_k = 1$ give, term by term,

$$\begin{aligned}
\bar{w}_{t+2} &= \sum_{k=1}^{K} \alpha_k \Big(w_t - 2\eta \, \nabla f + \eta^2 H_k \, \nabla f + O(\eta^3) \Big) \\
&= \Big(\underbrace{\textstyle\sum_k \alpha_k}_{=1} \Big) w_t - 2\eta \Big(\underbrace{\textstyle\sum_k \alpha_k}_{=1} \Big) \nabla f + \eta^2 \Big(\underbrace{\textstyle\sum_k \alpha_k H_k}_{=H} \Big) \nabla f + O(\eta^3) \\
&= w_t - 2\eta \, \nabla f + \eta^2 H \, \nabla f + O(\eta^3). \qquad (8)
\end{aligned}$$

Two facts make the averaging clean. First, w_t and ∇f are constants with respect to the index k (the global gradient $\nabla f = \sum_j \alpha_j g_j$ does not depend on the outer k), so they pass through the average unchanged via $\sum_k \alpha_k = 1$. Second, the only k-dependent object surviving to order η^2 is the Hessian H_k, and it is multiplied by the *k-independent* vector ∇f; hence the average of $H_k \, \nabla f$ is exactly $\big(\sum_k \alpha_k H_k\big) \nabla f = H \, \nabla f$, with no cross term. The $O(\eta^3)$ remainders, being a finite α_k-weighted ($\sum_k \alpha_k = 1$, $\alpha_k \geq 0$) average of $O(\eta^3)$ vectors with uniform constant, remain $O(\eta^3)$.

Step (v): compare to two centralized steps, and exhibit the cancellation. Two genuine centralized gradient-descent steps on f from w_t give, by the same Taylor expansion (1) applied to ∇f with global Hessian $H = \nabla^2 f(w_t)$,

$$\begin{aligned}
u_1 &= w_t - \eta \, \nabla f, \\
G^{(2)}(w_t) &= u_1 - \eta \, \nabla f(u_1) = u_1 - \eta \big(\nabla f - \eta H \, \nabla f + O(\eta^2) \big) \\
&= w_t - 2\eta \, \nabla f + \eta^2 H \, \nabla f + O(\eta^3), \qquad (9)
\end{aligned}$$

which is identical, term by term up to $O(\eta^3)$, to the corrected averaged round (8). Subtracting,

$$\bar{w}_{t+2} - G^{(2)}(w_t) = O(\eta^3). \qquad (10)$$

To see precisely what was cancelled, repeat Step (iv) for the *uncorrected* FedAvg endpoint $w^k_{(2),\text{FedAvg}} = w_t - 2\eta g_k + \eta^2 H_k g_k + O(\eta^3)$ of the deep dive. Averaging,

$$\bar{w}^{\text{FedAvg}}_{t+2} = w_t - 2\eta \sum_k \alpha_k g_k + \eta^2 \sum_k \alpha_k H_k g_k + O(\eta^3) = w_t - 2\eta \, \nabla f + \eta^2 \sum_k \alpha_k H_k g_k + O(\eta^3), \quad (11)$$

using $\sum_k \alpha_k g_k = \nabla f$. Subtracting the centralized expansion (9) from (11), the w_t and $-2\eta \, \nabla f$ terms cancel and the leading deviation is

$$\bar{w}^{\text{FedAvg}}_{t+2} - G^{(2)}(w_t) = \eta^2 \Big(\sum_{k=1}^{K} \alpha_k H_k g_k - H \, \nabla f \Big) + O(\eta^3) = \eta^2 \, \mathrm{Cov}_k(H_k, g_k) + O(\eta^3), \qquad (12)$$

which is exactly eq. (3.5) of the deep dive, by the definition of Cov_k recorded in the Working notation.

The cancellation is now transparent at the level of the order-η^2 coefficient. The uncorrected coefficient is $\sum_k \alpha_k H_k g_k$, in which the matrix H_k multiplies the *client-specific* gradient g_k; its average does *not* factor, leaving the residual $\sum_k \alpha_k H_k g_k - H \, \nabla f = \mathrm{Cov}_k(H_k, g_k)$, which is generically nonzero whenever the clients are heterogeneous (different H_k correlated with different g_k). The corrected coefficient, by contrast, is $\sum_k \alpha_k H_k \, \nabla f$, in which H_k multiplies the *global* gradient ∇f — a constant in k — so the average factors exactly as $\big(\sum_k \alpha_k H_k\big) \nabla f = H \, \nabla f$, leaving no residual. Equivalently, the difference between the corrected and uncorrected order-η^2 terms is

$$\eta^2 \sum_{k=1}^{K} \alpha_k H_k (\nabla f - g_k) = \eta^2 \Big(H \, \nabla f - \sum_{k=1}^{K} \alpha_k H_k g_k \Big) = -\, \eta^2 \, \mathrm{Cov}_k(H_k, g_k), \qquad (13)$$

i.e. the control variate replaces the local gradient g_k that H_k would have acted on by the global gradient ∇f inside the second step, and the resulting change is precisely $-\eta^2 \operatorname{Cov}_k(H_k, g_k)$, the exact negative of the FedAvg drift (12). Adding this to (12) sends the leading deviation to zero, leaving only $O(\eta^3)$, which is (10).

This proves both displayed equalities of Theorem 2: the corrected averaged round matches two centralized steps to $O(\eta^3)$, while the uncorrected round retains the $\Theta(\eta^2)$ heterogeneity term. The order-η^2 deviation $\eta^2 \operatorname{Cov}_k(H_k, g_k)$ therefore vanishes to first order under the control-variate correction. □

Discussion

What this gives us. The leading-order client-heterogeneity penalty of running more than one local step — the term $\eta^2 \operatorname{Cov}_k(H_k, g_k)$ that the deep dive identified as the entire $O(\eta^2)$ price of the second local step on non-IID data — is removed exactly by the control-variate correction $g_{\text{local}}(w) \mapsto g_{\text{local}}(w) - c_k + c$. After the fix, a corrected two-step federated round reproduces two centralized full-batch gradient-descent steps up to $O(\eta^3)$, i.e. the per-round progress is, to leading order, indistinguishable from centralized training *regardless of how heterogeneous the partition is.* The mechanism is explicit and local: $-c_k$ subtracts the client's own gradient and $+c$ inserts the global one, so each corrected step is centered on ∇f rather than on the client's biased g_k. This is the formal content behind importing SCAFFOLD-style control variates (Karimireddy et al. 2020) into FedAvg, and it is the leading-order counterpart of the proximal anchor used by FedProx (Li et al. 2020), which attacks the same drift object by a different route.

What this does not give us. The result is a *leading-order*, single-round, per-step statement, not a convergence theorem: it does not by itself bound the number of communication rounds, prove descent of f, or rule out divergence; it only equates one corrected two-step round with two centralized steps to $O(\eta^3)$. It says nothing about the $O(\eta^3)$ and higher terms, which still carry genuine client-dependence (third derivatives of F_k, the propagated remainder $\eta \cdot O(\eta^2)$ of Step (iii)); the correction zeroes the η^2 coefficient, not the whole gap. The analysis is deterministic and full-batch, with $g_{\text{local}} = \nabla F_k$ exact; it does not address stochastic minibatch noise (where c_k, c are themselves noisy estimates and the cancellation holds only in expectation), partial client participation, or the extra per-client state and the control-variate refresh that a multi-round deployment requires. The proof also assumes the standard FedAvg shared initialization $w^k_{(0)} = w_t$; without it even the first step (4) would differ across clients.

Where it would break. Two assumptions are load-bearing. (1) *Static, accurate control variates.* The cancellation in Steps (i)–(ii) relied twice on $c_k = g_k = \nabla F_k(w_t)$ holding for the whole round. If c_k is *stale* — estimated at an earlier global model $w_{t'} \neq w_t$, as happens with intermittent participation — then $c_k = \nabla F_k(w_{t'}) = g_k - H_k(w_t - w_{t'}) + \ldots$, and the leftover $-c_k + c$ no longer cancels g_k cleanly: a residual proportional to the staleness gap $w_t - w_{t'}$ re-enters at order η in the corrected direction and hence at order η^2 in the iterate, partially restoring a drift term. The cleaner $c_k = g_k$ the estimate, the more of the η^2 covariance is removed; an arbitrarily stale c_k recovers no benefit. (2) *Mild nonlinearity (validity of the quadratic model).* The whole argument truncates the Taylor expansion (1) at first order in the gradient (second order in F_k). Under strong nonlinearity — large third derivatives $\nabla^3 F_k$, or a learning rate η large enough that the round's displacements leave the neighborhood where (1) holds — the discarded $O(\eta^3)$ terms are no longer negligible and carry their own uncancelled client-heterogeneity contributions; the control variate is then guaranteed to remove only the η^2 piece, and the practical benefit shrinks as η grows or the local landscapes become more curved/heterogeneous at third order. Pushing the number of local steps well beyond two compounds this: each additional step adds higher-order remainders that the single first-order control variate does not address.

Empirical companion. This proof is the validation file for the improvement `heterogeneity-control-variate` in `05-improvements.tex` (subsection M.1). If a measurement-mode prototype also exists at `improvements/heterogeneity-control-variate.py`, it serves as a sanity check, not as evidence — the theorem above is the load-bearing claim.